

for $n \geq n_0$, $n \geq n_0 \Rightarrow x_n \in U$. Thus

$$\lim_{n \rightarrow \infty} x_n = x_0.$$



Recall that if X is an uncountable set then the cocountable topology & the discrete topology on X are different. But as we can see from the examples 2 & 3 given above the collections of convergent sequences in these two spaces are the same. Hence we conclude that the collection of convergent sequences in a topological space is insufficient in describing the topology of that space.

Another example regarding the insufficiency of ^(convergent) sequences is related to continuous functions:

Theorem: Let (X, τ) & (Y, ϑ) be two topological spaces & $f: X \rightarrow Y$ be a τ - ϑ continuous function. If $x_0 \in X$ & $\lim_{n \rightarrow \infty} x_n = x_0$ then

$$\lim_{n \rightarrow \infty} f(x_n) = f(x_0).$$

Proof: Exercise.

But the converse of this result need not be correct, in general:

Counterexample: Let $X = \mathbb{R}$ &

τ be the cocountable topology on X .

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by

$$f(x) = \begin{cases} \sqrt{2} & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \notin \mathbb{Q} \end{cases}$$

- $\{1\} \in \mathcal{T}_K$. But $f^{-1}(\{1\}) = \mathbb{R} \setminus \mathbb{Q}$ is not countable & hence $f^{-1}(\{1\}) \notin \mathcal{T}_K$.

Thus f is not $\mathcal{T}-\mathcal{T}$ continuous.

- Let $x_0 \in \mathbb{R}$ & let (x_n) be a sequence of real numbers s.t.

$$\lim_{n \rightarrow \infty} x_n = x_0. \quad \text{In this case}$$

$$\lim_{n \rightarrow \infty} f(x_n) = f(x_0) \quad (\text{WHY?}).$$

In other words in a topological space continuity implies sequential continuity but sequential continuity need not imply continuity.

Bases and subbases

Definition: Let $(X, \mathcal{T})$ be a top. sp. A class $\mathcal{B}$ of $\mathcal{T}$ -open subsets of X , i.e. $\mathcal{B} \subseteq \mathcal{T}$, is a base (or a basis) for $\mathcal{T}$ iff every open set $U \in \mathcal{T}$ is the union of members of $\mathcal{B}$.

Proposition: Let $(X, \mathcal{T})$ be a top. sp. & $\mathcal{B} \subseteq \mathcal{T}$. Then $\mathcal{B}$ is a base for $\mathcal{T} \iff$ for every $U \in \mathcal{T}$ & every $x \in U$ there exists a $B \in \mathcal{B}$ s.t. $x \in B \subseteq U$.

Proof:

$\Rightarrow$ Let $U \in \mathcal{T}$ and let $x \in U$. Because $\mathcal{B}$ is a base for $\mathcal{T}$ & $U \in \mathcal{T}$ there exists a $\mathcal{B}' \subseteq \mathcal{B}$ s.t. $U = \bigcup \mathcal{B}'$. On the other

hand $x \in U = \bigcup \mathcal{B}'$ & so there
exists a $B \in \mathcal{B}' \subseteq \mathcal{B}$ s.t.
 $x \in B \subseteq U$.

$\Leftarrow$ Let $U \in \mathcal{T}$. Take any point x in U .
By hypothesis there exists a $B_x \in \mathcal{B}$
s.t. $x \in B_x \subseteq U$. Define

$$\mathcal{B}'_x = \{ B_x \mid B_x \in \mathcal{B}, x \in B_x \subseteq U \} \subseteq \mathcal{B}.$$

Then

$$U \subseteq \bigcup_{x \in U} \mathcal{B}'_x = \bigcup_{B_x \in \mathcal{B}'_x} B_x \subseteq U$$

i.e. $U = \bigcup \mathcal{B}'_x$

Thus $\mathcal{B}$ is a base for $\mathcal{T}$.

Example: Let $(X, \mathcal{D})$ be a discrete space. Show that the class

$\mathcal{B} = \{\{x\} \mid x \in X\}$ is a base for $\mathcal{D}$.

Solution:

• $\mathcal{B} \subseteq \mathcal{D}$:

$\mathcal{D}$ is the discrete topology on X & hence any subset of X is $\mathcal{D}$ -open. In particular $\{x\} \in \mathcal{D}$ for each $x \in X$.

$\therefore \mathcal{B} \subseteq \mathcal{D}$.

• Let $A \in \mathcal{D}$. Because we can express A as $A = \bigcup_{a \in A} \{a\}$

any $\mathcal{D}$ -open subset of X can be expressed as a union of members of $\mathcal{B}$.

$\therefore \mathcal{B}$ is a base for $\mathcal{D}$.

Example: Let $X = \{a, b, c\}$. Can the class $\mathcal{B} = \{\{a, b\}, \{b, c\}\}$ be a base for a topology on X ?

Solution: No. Because if it were a base for a topology $\mathcal{T}$ on X then $\{a, b\}, \{b, c\} \in \mathcal{T}$. Because $\mathcal{T}$ is a top. on X

$\{a, b\} \cap \{b, c\} = \{b\} \in \mathcal{T}$ by (T2).

But then we would be able to express $\{b\}$ as a union of members of $\mathcal{B}$ which is impossible.

Now the following question arises;

Suppose we have an arbitrary class $\mathcal{B}$ of subsets of a nonempty set X .

When will the class $\mathcal{B}$ be a base for some topology on X ? Since

X is open in every topology on X , clearly $X = \bigcup \mathcal{B} = \bigcup \{B \mid B \in \mathcal{B}\}$.

The following theorem gives necessary and sufficient